

Project Title

Full Stack Expense Tracker Application

1. Project Overview

This project is a **Full Stack Expense Tracker Application** developed using **React.js, Node.js, Express.js, and MongoDB**. The application helps users manage their daily income and expenses by recording transactions, categorizing spending, applying filters, and visualizing financial data using charts and graphs. It demonstrates practical implementation of authentication, CRUD operations, and data visualization in a MERN stack environment.

2. Objectives of the Project

- To build a real-world personal finance management system
 - To implement secure user authentication using JWT
 - To perform CRUD operations for income and expense transactions
 - To categorize and filter financial data efficiently
 - To visualize expenses using charts and graphs
 - To follow industry-standard project structure and coding practices
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3. Technology Stack

Frontend

- React.js
- Context API (Auth & Transaction Management)
- Chart Library (Chart.js / Recharts)
- CSS

Backend

- Node.js
- Express.js
- RESTful APIs

Database

- MongoDB
 - Mongoose
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4. Project Architecture

The application follows a **Client–Server Architecture**:

- Frontend (React) handles user interface and state management
- Backend (Node + Express) manages authentication and APIs
- MongoDB stores users and transaction data securely

5. Core Features

- JWT-based user authentication (Login / Register)
- Add, edit, and delete income & expense transactions
- Category-wise expense management (Food, Rent, Travel, Shopping, Salary, etc.)
- Filter transactions by date, category, and type
- Graphical analysis using charts (monthly trends, category-wise pie charts)

6. Functional Modules

6.1 User Authentication Module

- User registration and login
- Secure JWT token-based authentication
- Protected routes

6.2 Transaction Management Module

- Add income and expense records
- Edit and delete transactions
- Categorization of transactions

6.3 Filter & Analytics Module

- Filter by date range, category, and transaction type
- Monthly expense summary
- Visual charts and graphs

7. Project Directory Structure

Backend Structure

```
expense-tracker/
├── backend/
│   ├── src/
│   │   ├── config/
│   │   │   └── db.js
│   │   ├── models/
│   │   │   ├── User.js
│   │   │   └── Transaction.js
│   │   ├── controllers/
│   │   │   ├── authController.js
│   │   │   └── transactionController.js
│   │   ├── routes/
│   │   │   ├── authRoutes.js
│   │   │   └── transactionRoutes.js
│   │   ├── middleware/
│   │   │   ├── authMiddleware.js
│   │   │   └── errorMiddleware.js
│   │   └── server.js
│   ├── package.json
│   └── .env
```

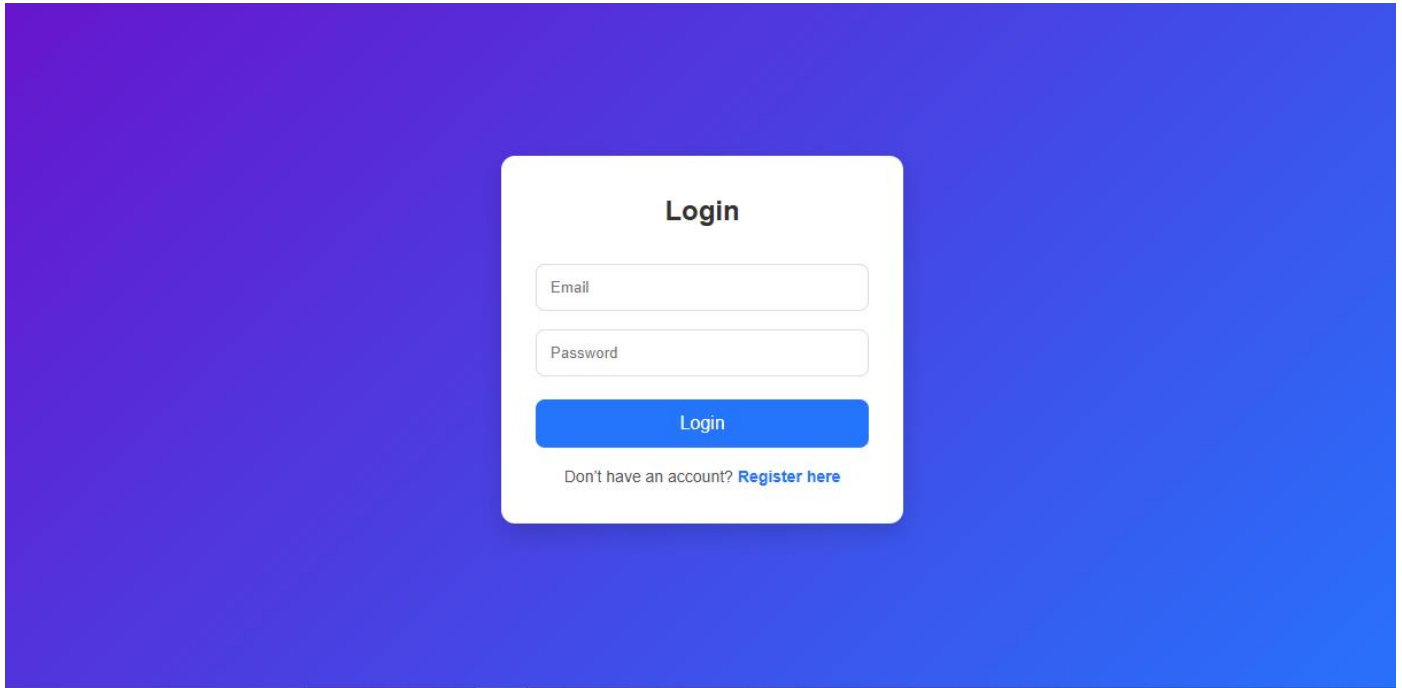
Frontend Structure

```
frontend/
├── src/
│   ├── api/
│   │   ├── authApi.js
│   │   └── transactionApi.js
│   ├── components/
│   │   ├── Navbar.jsx
│   │   ├── TransactionForm.jsx
│   │   ├── TransactionList.jsx
│   │   └── ChartSection.jsx
│   ├── context/
│   │   ├── AuthContext.jsx
│   │   └── TransactionContext.jsx
│   ├── pages/
│   │   ├── Login.jsx
│   │   ├── Register.jsx
│   │   └── Dashboard.jsx
│   ├── styles/
│   │   └── app.css
│   ├── App.jsx
│   └── main.jsx
└── package.json
```

9. User Interface Description

Login & Register Page

- Secure user authentication forms

A login form UI mockup centered on a blue gradient background. The form is a white rounded rectangle with a shadow. It has a title 'Login' in bold black text. Below the title are two input fields: 'Email' and 'Password', both with light gray placeholder text. Below the input fields is a blue button with the text 'Login' in white. At the bottom of the form, there is a link that says 'Don't have an account? Register here' in a smaller, lighter blue font.

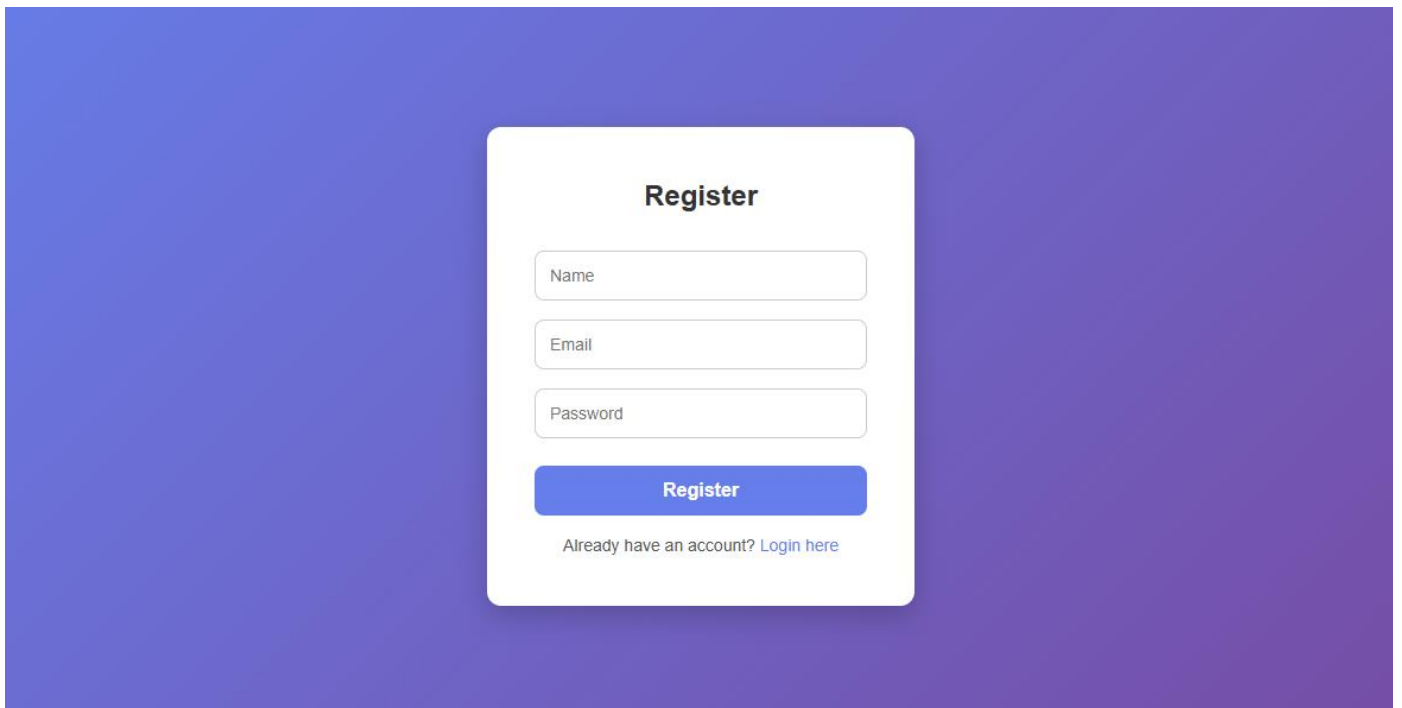
Login

Email

Password

Login

Don't have an account? [Register here](#)

A register form UI mockup centered on a purple gradient background. The form is a white rounded rectangle with a shadow. It has a title 'Register' in bold black text. Below the title are three input fields: 'Name', 'Email', and 'Password', all with light gray placeholder text. Below the input fields is a blue button with the text 'Register' in white. At the bottom of the form, there is a link that says 'Already have an account? Login here' in a smaller, lighter blue font.

Register

Name

Email

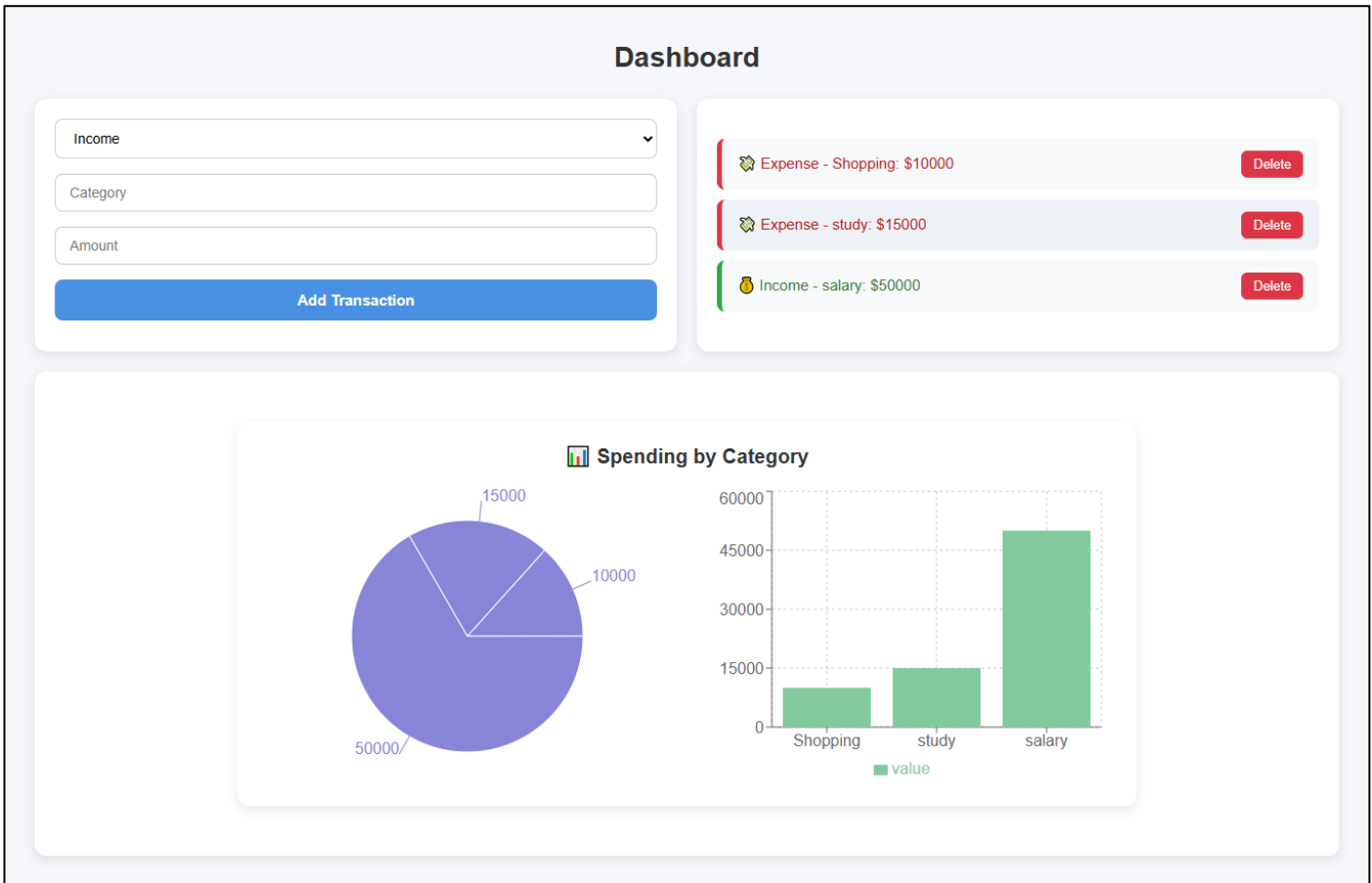
Password

Register

Already have an account? [Login here](#)

Dashboard Page

- Overview of income and expenses
- Charts and graphs for financial analysis



10. Security & Validation

- JWT-based authentication
- Protected API routes
- Input validation on frontend & backend
- Environment variables for sensitive credentials